Projet électronique séance 3

Table 5. Debug connector CN4 (SWD)

Pin	CN4	Designation
1	VDD_TARGET	VDD from application
2	SWCLK	SWD clock
3	GND	ground
4	SWDIO	SWD data input/output
5	NRST	RESET of target STM32
6	SWO	Reserved

Figure 8. Using ST-LINK/V2-1 to program the STM32 on an external application

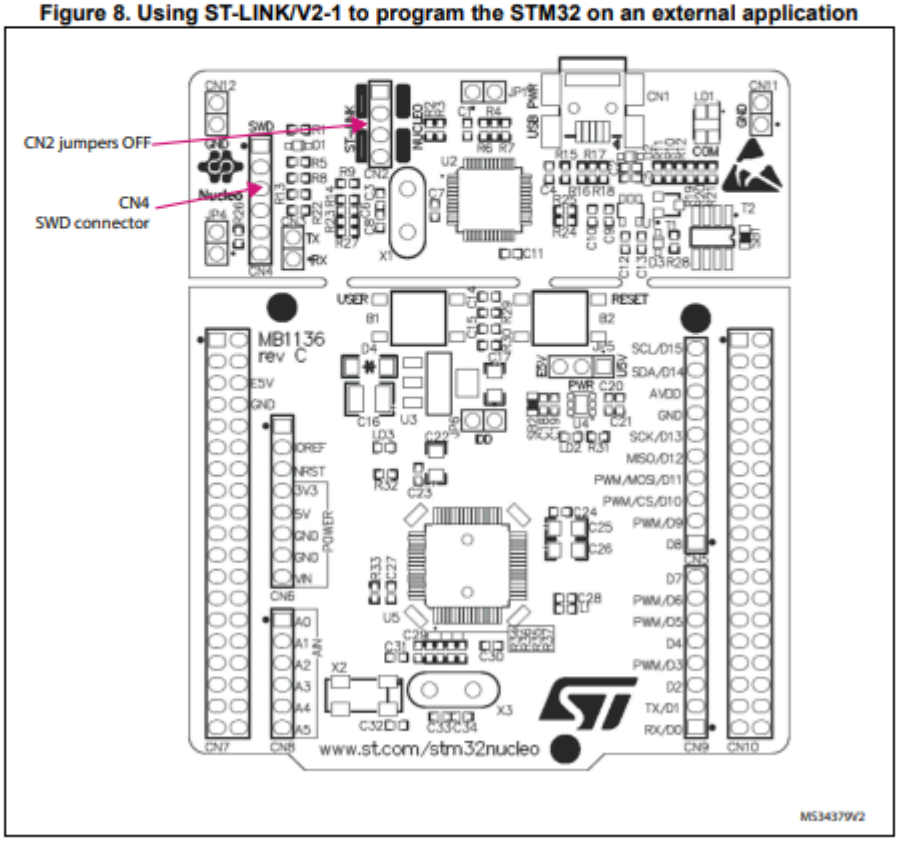

tableau de connection pour la programmation/débogage

nomsConnections capteur-STM :

Datasheet - STM32G431x6 STM32G431x8 STM32G431xB - Arm® Cortex®... 53 / 198 74%

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Table 12. STM32G431x6/x8/xB pin definition[1] (continued)

Pin Number									Pin name (function after reset)	Pin type	I/O structure	Notes	Alternate function	Additional functions
UFQFPN32	LQFP32	UFQFPN48	LQFP48	WLCSP49	LQFP64	UFBGA64	LQFP80	LQFP100						
9	9	12	12	E5	18	D4	18	26	PA4	I/O	TT_a	-	TIM3_CH2, SPI1_NSS, SPI3_NSS/ I2S3_WS,USART2_CK, SAI1_FS_B, EVENTOUT	ADC2_IN17, DAC1_OUT1, COMP1_INM
10	10	13	13	F5	19	E4	19	27	PA5	I/O	TT_a	-	TIM2_CH1, TIM2_ETR, SPI1_SCK, UCPD1_FRSTX, EVENTOUT	ADC2_IN13, DAC1_OUT2, COMP2_INM, OPAMP2_VINM
11	11	14	14	G6	20	G3	20	28	PA6	I/O	TT_a	-	TIM16_CH1, TIM3_CH1,TIM8_BKIN, SPI1_MISO, TIM1_BKIN, COMP1_OUT, LPUART1_CTS, EVENTOUT	ADC2_IN3, OPAMP2_VOUT
12	12	15	15	D5	21	H3	21	29	PA7	I/O	TT_a	-	TIM17_CH1, TIM3_CH2, TIM8_CH1N, SPI1_MOSI, TIM1_CH1N, COMP2_OUT, UCPD1_FRSTX, EVENTOUT	ADC2_IN4, COMP2_INP, OPAMP1_VINP, OPAMP2_VINP
-	-	16	-	D4	22	D5	22	30	PC4	I/O	FT_la	-	TIM1_ETR, I2C2_SCL, USART1_TX, EVENTOUT	ADC2_IN5
-	-	-	-	-	23	F4	23	31	PC5	I/O	TT_a	-	TIM15_BKIN, SAI1_D3, TIM1_CH4N, USART1_RX, EVENTOUT	ADC2_IN11, OPAMP1_VINM, OPAMP2_VINM, WKUP5
13	13	17	16	G5	24	E5	24	32	PB0	I/O	TT_a	-	TIM3_CH3, TIM8_CH2N, TIM1_CH2N, UCPD1_FRSTX, EVENTOUT	ADC1_IN15, COMP4_INP, OPAMP2_VINP, OPAMP3_VINP
-	-	18	17	E4	25	F5	25	33	PB1	I/O	TT_a	-	TIM3_CH4, TIM8_CH3N, TIM1_CH3N, COMP4_OUT, LPUART1_RTS_DE, EVENTOUT	ADC1_IN12, COMP1_INP, OPAMP3_VOUT
-	-	19	18	F4	26	H4	26	34	PB2	I/O	TT_a	-	RTC_OUT2, LPTIM1_OUT, I2C3_SMBA, EVENTOUT	ADC2_IN12, COMP4_INM, OPAMP3_VINM

Datasheet - STM32G431x6 STM32G431x8 STM32G431xB - Arm® Cortex®... 54 / 198 74%

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Table 12. STM32G431x6/x8/xB pin definition[1] (continued)

Pin Number									Pin name (function after reset)	Pin type	I/O structure	Notes	Alternate function	Additional functions
UFQFPN32	LQFP32	UFQFPN48	LQFP48	WLCSP49	LQFP64	UFBGA64	LQFP80	LQFP100						
14	14	-	19	G4	27	G4	27	35	VSSA	S	-	-	-	-
-	-	20	20	F3	28	G5	28	36	VREF+	S	-	-	-	VREFBUF_OUT
-	-	21	21	G3	29	H5	29	37	VDDA	S	-	-	-	-
15	15	-	-	-	-	-	-	-	VDDA/VREF+	S	-	-	-	-
-	-	-	-	-	-	-	30	38	PE7	I/O	TT_a	-	TIM1_ETR, SAI1_SD_B, EVENTOUT	COMP4_INP
-	-	-	-	-	-	-	31	39	PE8	I/O	FT_a	-	TIM1_CH1N, SAI1_SCK_B, EVENTOUT	COMP4_INM
-	-	-	-	-	-	-	32	40	PE9	I/O	FT	-	TIM1_CH1, SAI1_FS_B, EVENTOUT	-
-	-	-	-	-	-	-	33	41	PE10	I/O	FT	-	TIM1_CH2N, SAI1_MCLK_B, EVENTOUT	-
-	-	-	-	-	-	-	34	42	PE11	I/O	FT	-	TIM1_CH2, EVENTOUT	-
-	-	-	-	-	-	-	35	43	PE12	I/O	FT	-	TIM1_CH3N, EVENTOUT	-
-	-	-	-	-	-	-	36	44	PE13	I/O	FT	-	TIM1_CH3, EVENTOUT	-
-	-	-	-	-	-	-	37	45	PE14	I/O	FT	-	TIM1_CH4, TIM1_BKIN2, EVENTOUT	-
-	-	-	-	-	-	-	38	46	PE15	I/O	FT	-	TIM1_BKIN, TIM1_CH4N, USART3_RX, EVENTOUT	-
-	-	22	22	F2	30	H6	39	47	PB10	I/O	TT_a	-	TIM2_CH3, USART3_TX, LPUART1_RX, TIM1_BKIN, SAI1_SCK_A, EVENTOUT	OPAMP3_VINM
16	16	-	23	G2	31	G7	40	48	VSS	S	-	-	-	-
17	17	23	24	G1	32	H8	41	49	VDD	S	-	-	-	-
-	-	24	25	E3	33	H7	42	50	PB11	I/O	FT_a	-	TIM2_CH4, USART3_RX, LPUART1_TX, EVENTOUT	ADC12_IN14

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Table 12. STM32G431x6/x8/xB pin definition[1] (continued)

Pin Number									Pin name (function after reset)	Pin type	I/O structure	Notes	Alternate function	Additional functions
UFQFPN32	LQFP32	UFQFPN48	LQFP48	WLCSP49	LQFP64	UFBGA64	LQFP80	LQFP100						
22	22	34	34	B1	46	B8	60	73	PA12	I/O	FT_u	-	TIM16_CH1, I2SCKIN, TIM1_CH2N, USART1_RTS_DE, COMP2_OUT, FDCAN1_TX, TIM4_CH2, TIM1_ETR, EVENTOUT	USB_DP
-	-	-	35	-	47	B7	61	74	VSS	S	-	-	-	-
-	-	35	36	-	48	A8	62	75	VDD	S	-	-	-	-
23	23	36	37	B2	49	C7	63	76	PA13	I/O	FT_f	(5)	SWDIO-JTMS, TIM16_CH1N, I2C1_SCL, IR_OUT, USART3_CTS, TIM4_CH3, SAI1_SD_B, EVENTOUT	-
24	24	37	38	B3	50	C6	64	77	PA14	I/O	FT_f	(5)	SWCLK-JTCK, LPTIM1_OUT, I2C1_SDA, TIM8_CH2, TIM1_BKIN, USART2_TX, SAI1_FS_B, EVENTOUT	-
25	25	38	39	A1	51	A7	65	78	PA15	I/O	FT_f	(5)	JTDI, TIM2_CH1, TIM8_CH1, I2C1_SCL, SPI1_NSS, SPI3_NSS/ I2S3_WS, USART2_RX, UART4_RTS_DE, TIM1_BKIN, TIM2_ETR, EVENTOUT	-
-	-	39	-	-	52	C5	66	79	PC10	I/O	FT	-	TIM8_CH1N, UART4_TX, SPI3_SCK/ I2S3_CK, USART3_TX, EVENTOUT	-
-	-	40	-	A2	53	B6	67	80	PC11	I/O	FT_f	-	TIM8_CH2N, UART4_RX, SPI3_MISO, USART3_RX, I2C3_SDA, EVENTOUT	-

Table 12. STM32G431x6/x8/xB pin definition[1] (continued)

Pin Number									Pin name (function after reset)	Pin type	I/O structure	Notes	Alternate function	Additional functions
UFQFPN32	LQFP32	UFQFPN48	LQFP48	WLCSP49	LQFP64	UFBGA64	LQFP80	LQFP100						
27	27	42	41	B4	57	C4	73	91	PB4	I/O	FT_c	(5) (6)	JTRST, TIM16_CH1, TIM3_CH1, TIM8_CH2N, SPI1_MISO, SPI3_MISO, USART2_RX, TIM17_BKIN, SAI1_MCLK_B, EVENTOUT	UCPD1_CC2
28	28	43	42	A43	58	B4	74	92	PB5	I/O	FT_f	-	TIM16_BKIN, TIM3_CH2, TIM8_CH3N, I2C1_SMBA, SPI1_MOSI, SPI3_MOSI/ I2S3_SD, USART2_CK, I2C3_SDA, TIM17_CH1, LPTIM1_IN1, SAI1_SD_B, EVENTOUT	-
29	29	44	43	C4	59	A4	75	93	PB6	I/O	FT_c	(6)	TIM16_CH1N, TIM4_CH1, TIM8_CH1, TIM8_ETR, USART1_TX, COMP4_OUT, TIM8_BKIN2, LPTIM1_ETR, SAI1_FS_B, EVENTOUT	UCPD1_CC1
30	30	45	44	A5	60	A3	76	94	PB7	I/O	FT_f	-	TIM17_CH1N, TIM4_CH2, I2C1_SDA, TIM8_BKIN, USART1_RX, COMP3_OUT, TIM3_CH4, LPTIM1_IN2, UART4_CTS, EVENTOUT	PVD_IN
31	31	46	45	B5	61	B3	77	95	PB8-BOOT0	I/O	FT_f	(7)	TIM16_CH1, TIM4_CH3, SAI1_CK1, I2C1_SCL, USART3_RX, COMP1_OUT, FDCAN1_RX, TIM8_CH2, TIM1_BKIN, SAI1_MCLK_A, EVENTOUT	-

connexions EEPROM-STM32

Si tu avais une EEPROM SPI comme la M95512-RMN6TP

Les connexions seraient :

EEPROM	STM32
SCK	SPI_SCK
SI	SPI_MOSI
SO	SPI_MISO
CS	GPIO (Chip Select)

boardhttps://www.st.com/resource/en/application_note/an5717-how-to-setup-and-run-the-vl53l5cxsatelusing-an-stm32-nucleo64-board-stmicroelectronics.pdf

How to use the VL53L5CX with STMicroelectronics' X-CUBE-TOF1 Time-of-Flight sensor software packages for STM32CubeMX:

https://www.st.com/resource/en/user_manual/um2974-how-to-use-the-vl53l5cx-with-stmicroelectronics-xcubetof1-timeofflight-sensor-software-packages-for-stm32cubemx-stmicroelectronics.pdf